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CS270

13 April 2026

Ontology of End-of-Life Care Preferences

1. Application Area

Advance directives (ADs) are legal documents in which patients specify their end-of-life (EOL) care preferences. Despite their importance, current advance directives are almost entirely free-text documents that cannot be queried, reasoned over, or transferred reliably across care settings. Clinicians frequently encounter ADs at moments of acute crisis and must interpret ambiguous natural language under time pressure—a known failure mode in critical care.

The core computational problem is the absence of a computable, machine-actionable representation of EOL preferences. Existing standards address data transport (e.g., the HL7 FHIR Advance Directive Interoperability implementation guide) but not semantic reasoning: they specify how to store and transmit a directive, not how to infer what care a patient would want given their current clinical state.

This project proposes to build an OWL ontology of EOL care preferences that enables a reasoner to match a patient's documented preferences to their current clinical situation and infer appropriate care decisions. We scope our work to one disease trajectory (advanced heart failure) to ensure tractability within the quarter, while designing the ontology to be generalizable.

2. Methodology

Although there do not exist datasets with raw advanced directives, advanced directive templates and related datasets will be used to inform our ontology. For example, we will use state advance directive templates, as well as the POLST dataset, which contains structured, clinician-validated fields capturing patient preferences on life-sustaining treatments such as CPR status, level of medical intervention (comfort-focused vs. full treatment), and artificial nutrition. We will also review published clinical vignettes in literature in order to understand real-world examples of when advance directives can present ethical dilemmas and challenges. This ensures that we ground our ontology in realistic, standardized representations of end-of-life decision-making.

Drawing on these resources, we will generate a structured vocabulary for the ontology, with top-level classes anticipated to be preference statement, clinical condition, intervention, decision-maker, and context. We will ensure that conditional preferences (if-then structure), negation, and value statements will be documented explicitly. We will build the ontology in OWL and Protégé, and will use Python to programmatically instantiate ontology individuals from structured directive input.

3. Anticipated Challenges

3.1 Negation in OWL

Exclusionary preferences (“I do not want mechanical ventilation”) are non-trivial to represent under OWL’s open-world assumption. We will address this with a deliberate design choice: a `hasPreference` object property paired with a boolean `isNegated` data property, and where necessary, `complementOf` class expressions. We will document this decision and analyze where it breaks down.

3.2 Conditional and Temporal Preferences

Patient preferences are frequently conditional (“only if there is reasonable hope of recovery”) and time-sensitive (preferences change as disease progresses). OWL does not natively support temporal reasoning. We will model condition-contingency using `hasActivationCondition` restrictions and acknowledge temporal dynamics as a limitation, suggesting an external temporal reasoner as future work.

3.3 Vagueness as a First-Class Problem

Terms like “heroic measures” or “reasonable quality of life” are intentionally vague. Rather than forcing false precision, we will model a `VaguePreference` class that preserves the original language as a data property while assigning it to the nearest formal class. This allows the ontology to acknowledge what it cannot fully represent.

3.4 Scope Management

End-of-life care intersects law, ethics, culture, religion, and medicine. We will strictly scope to advanced heart failure as the disease context, set aside cross-jurisdictional legal variation, and defer a full user study with clinicians to future work, relying instead on the self-contained coverage analysis and scenario tests for evaluation.

Timeline

Week 3–4 (by April 27)

- Collect and annotate 30-40 advanced directive and POLST documents

- Produce concept inventory (raw vocabulary)
- Draft top-level class hierarchy
- Begin Protege implementation of core class skeleton

Week 5–6 (Progress Report: May 6)

- Complete OWL ontology with full class hierarchy, object/data properties, and restrictions
- Ground clinical concepts in SNOMED CT using OwlReady2 and Python
- Implement DL query generation pipeline
- Begin coverage analysis on 25-30 preference statements

Week 7–8 (by May 20)

- Complete coverage analysis and categorize failures
- Design and run 15 clinical vignette scenario tests
- Begin expressivity comparison against FHIR ADI and POLST

Week 9–10 (Presentations: June 1 | Final Report: June 5)

- Finalize analysis and synthesize findings
- Prepare poster and 8-10 minute presentation
- Write final report (10 pages): motivation, design decisions, pipeline, evaluation, discussion